

Platform Guide

Code & Functionality

A smart-irrigation and crop-health platform for India's small farms, combining a solar IoT field node, satellite and weather data, verified agronomy and AI that explains the numbers.

Every drop measured. Every field protected.

TEAM

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CODE

github.com/marvillage/AgriGuard

EVENT

EcoLogic 1.0 Sustainability Hackathon (Mercer | Mettl) · Smart Agriculture

VERSION

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How to read this guide. Chapters 1–7 explain what AgriGuard does and why each number can be trusted. Chapters 8–14 are the technical reference for developers. Every statistic carries its source. Results from simulations are marked **SIMULATED** and results from real runs are marked **MEASURED**.

1 Overview

Most irrigation pumps in India run on a timer, a habit or a hunch. Nobody can see how wet the root zone is, whether rain is coming, or how much water, electricity and fertilizer each decision costs. AgriGuard makes that visible and acts on it.

The problem in numbers

247.22 BCM

groundwater extracted in India in 2025; stage of extraction 60.63%

CGWB, Dynamic Ground Water Resources of India 2025

10.8%

of groundwater assessment units are over-exploited

CGWB 2025

16.9%

of India's electricity is used by agriculture (FY 2022–23)

CEA General Review 2024

86.08%

of holdings are small or marginal (under 2 ha); average 1.08 ha

Agriculture Census 2015–16

Also: only 30–50% of flood-irrigation water is used by the crop (ICAR), and about 70% of the world's freshwater withdrawals go to agriculture (FAO).

What AgriGuard does: measure, decide, act, prove

1 • Measure

A solar-powered ESP32 field node reads soil moisture, soil and air temperature, humidity, NPK, water flow, energy, tank level and solar input every 60 seconds. Fields without a node use Open-Meteo's modelled soil moisture for their exact location, clearly labelled as an estimate. Sentinel-2 satellite images give crop greenness (NDVI).

2 • Decide

An FAO-56 root-zone water balance decides *water / wait / skip* for each field: skip when the soil is still wet or enough rain is likely, wait for the solar window if the farm has panels, and never run a dry tank. Risk scores, soil-test fertilizer plans and a leaf-disease scanner cover the rest of the crop.

3 • Act

In automatic mode the node switches the pump starter through a relay, with local safety cut-offs. The farmer sees one clear instruction in their own language and can ask "Why?". Gemini explains the real numbers in plain words and never changes them.

4 • Prove

Every litre, kWh, rupee and kg of CO₂ goes into a hash-chained savings ledger that shows whether each number was measured or estimated. Savings are counted only when real pump use is recorded. Control-plot trials compare a smart field against the usual practice.

At a glance

Area	What is in the product today
Languages	English, Hindi, Marathi, Punjabi, Telugu and Tamil across the whole app, with voice input and read-aloud
Crops	Irrigation and fertilizer profiles for 14 crops from FAO-56 and PAU/ICAR (wheat, paddy, maize, cotton, sugarcane, tomato, potato, onion, chilli, soybean, groundnut, mustard, chickpea, grapes); disease scanning for 38 crop/disease classes across 14 crops

Area	What is in the product today
Sensors	8 sensor types in the firmware; basic node $\approx$ ₹4,000–4,500, full node about ₹10,000 (itemised parts list, Indian retail 2026)
AI	On-device MobileNetV2 disease model; Gemini Flash models for explanations, second opinions and Soil Health Card reading; local Qwen2.5-VL as an offline fallback
Quality	38/38 automated tests; disease model tested on 236 real field photos; past-season replay on real 2025 weather
Deploy	Render Blueprint (PostgreSQL + API + web app), public code on GitHub

2 Who uses AgriGuard

Role	Can do
Farmer (FARMER)	Create farms and fields, register field nodes, control pumps, scan crops, plan fertilizer, chat with the AI copilot, see savings and run field trials on their own farms.
Agronomist (AGRONOMIST)	Join a farmer's farm with the farm's share code, watch risk across all joined farms on the Agronomist Desk, and send notes that arrive in the farmer's recommendations. Advisors read, the farmer stays in control.
FPO / admin (ADMIN)	See every farm, the way a Farmer Producer Organisation supports many members.

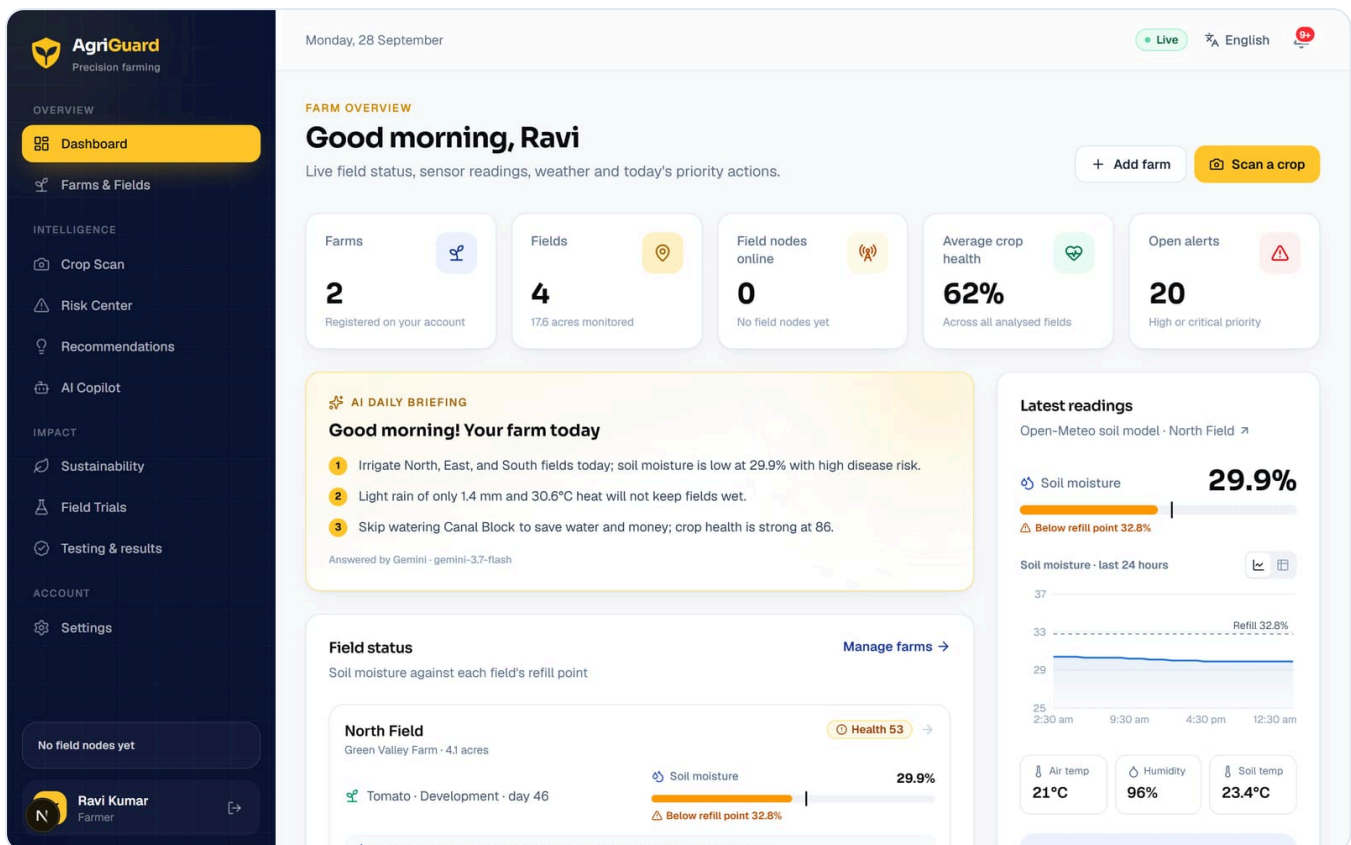
Demo accounts

All use the password `agriguard123` and real farm locations with real weather.

Email	Role	Language	Farms (state, electricity tariff)
demo@agriguard.in	Farmer	English	Green Valley Farm, Nashik, Maharashtra (MSEDCL ₹4.02/kWh) and Sunrise Acres, Ludhiana, Punjab (PSPCL ₹5.38/kWh)
sunita@agriguard.in	Farmer	Hindi	Narmada Kisan Farm, Dewas, Madhya Pradesh (MPERC ₹6.36/kWh)
anil@agriguard.in	Farmer	Marathi	Pawar Farms, Baramati, Maharashtra (MSEDCL ₹4.02/kWh)
lakshmi@agriguard.in	Farmer	Telugu	Krishna Delta Farm, Guntur, Andhra Pradesh (APEREC ₹8.13/kWh)
advisor@agriguard.in	Agronomist	English	Joined to Green Valley, Narmada Kisan and Krishna Delta farms
fpo@agriguard.in	Admin	English	All farms

Tariffs come from the state tariff orders cited in `backend/scripts/seed-demo.ts`. Where power is free or flat-rate for farmers (Punjab, Andhra Pradesh, Madhya Pradesh), the figure is the tariff or subsidy paid per kWh, so savings show the real cost to the system.

3 Feature guide, page by page



Dashboard: farm stats, the AI daily briefing, field status against each refill point, and the latest readings.

Dashboard

- Stat tiles: farms, fields and acres, field nodes online, average crop health, open high-priority alerts.
- **AI daily briefing:** three bullets (most urgent action, weather effect, one saving), generated from each field's live decision and risks, in the farmer's language; if no AI is available the rules engine writes it.
- **Field status:** soil moisture against the refill point for every field, with today's decision.
- **Latest readings:** 24-hour soil-moisture chart with the refill line, air and soil temperature, humidity; the source (field node or Open-Meteo model) is always shown.
- Weather, impact strip, priority recommendations, and onboarding for new users.

Farms & Fields

- Farm: name, location (search or map pin), irrigation method, electricity tariff, solar capacity (kW), and optionally the farmer's own usual watering depth and interval.
- Field: area, soil type (7 FAO-56 soil profiles), crop and planting date, irrigation method, pump flow and power, and an optional boundary drawn on the map (area computed from the polygon).

Field page (8 tabs)

Tab	What it shows
Overview	The irrigation decision card (action, litres, pump time, depth, method, ET _o /ETc, NDVI and solar-window chips, "Why? Ask AI"), live readings, the water budget, risk scores (crop health, water, disease, weather, nutrient, with "Why?"), weather strip and a soil-moisture forecast.
Pump & water	Auto / manual on / manual off pump control, smart schedules (time windows), irrigation log, decision log and the solar window.
Devices	Register a field node, copy or rotate its device key, and set up the tank (height, capacity, dry-run level) and which sensors are fitted.
Crop	Crop timeline by FAO-56 stage (initial, development, mid, late) with Kc, root depth and expected harvest; record harvests.
Map	Field boundary, Sentinel-2 NDVI overlay with opacity control, scene picker, zone shares (below 0.3 / 0.3–0.6 / above 0.6) and "Why?".
Fertilizer	Soil-test calculator: typed values, a Soil Health Card photo read by AI, or the NPK sensor. It gives urea/DAP/MOP amounts vs the blanket dose, a 3-split schedule, cost and saving; plans can be marked applied (recorded in the ledger) and explained with "Why?".
Readings	Reading history, raw sensor data and manual entry.
Report	Weekly field report: 7-day statistics computed by code, summarised by AI in 4 sections.

IRRIGATION DECISION Irrigate

Soil is below the refill point and no rain is due: irrigating 2,21,228 L.

ET_o 4.4 mm/day ETc 3.6 mm/day NDVI 0.60 • 10 Sept

Based on Open-Meteo's modelled soil moisture for this location, not a field sensor.
Decided 6 min ago

Why? Ask AI

Your soil moisture is at 29.9%, which is below your refill point of 32.8%. The rain expected in the next 24 hours is only 1.4 mm, which is not enough to skip watering. You should irrigate 2,21,228 litres of water using your drip system. Run your drip irrigation at 180 L/min for about 20.5 h.

AI explains the numbers above in simple words. The numbers come from AgriGuard's formulas, not from the AI.

Answered by Gemini · gemini-3.6-flash

PLANNED IRRIGATION

2.2 lakh L

Pump time	20.5 h	Depth	13.3 mm gross · 12 mm net
Method	Drip · 90% efficient	Pump flow	180 L/min

Irrigation decision with the "Why?" explanation. The numbers come from the FAO-56 engine; Gemini only puts them into words.

Crop Scan

- **Disease mode:** the photo is normalised, classified on-device by MobileNetV2 (38 PlantVillage classes), filtered to the crop the farmer selects, and flagged "low confidence" below 0.6. Pixel colours measure the discoloured leaf area.
- **AI second opinion:** Gemini looks at the same photo, gives its own diagnosis, says whether it agrees, and estimates the damaged area (shown next to the measured figure).
- **Verified advice:** symptoms, actions and prevention from the checked knowledge base in all six languages, plus "Why?", which relates the listed actions to the coming weather.

- **Pest mode:** Gemini identifies pests and pest damage.

 **AI second opinion**
Agrees with the model

Early Blight

AI confidence 98%

AI estimate of damaged area About 45% of the leaf

The leaf shows characteristic necrotic brown lesions with prominent concentric rings (target-board pattern) surrounded by chlorotic yellow halos. This fungal disease primarily attacks older foliage and spreads upwards during warm, humid conditions.

Suggested steps

- 1 Prune and destroy heavily infected lower leaves to minimize inoculum and improve air circulation.
- 2 Apply a registered protectant or systemic fungicide containing active ingredients like Mancozeb, Chlorothalonil, or Azoxystrobin, strictly following label directions.
- 3 Consult your local Krishi Vigyan Kendra (KVK) or extension officer to confirm recommended spray schedules and resistance management practices.

 Gemini · gemini-3.8-flash

A real tomato early-blight photo: the AI second opinion agrees with the on-device model and estimates the damaged area next to the measured one.

Risk Center, Recommendations and AI Copilot

- **Risk Center:** water, disease, weather and nutrient risk for every field on one page.
- **Recommendations:** prioritised actions (critical to low) from the engine, scans, devices and agronomists; mark done or dismissed; "Explain" and "Translate" with AI.
- **AI Copilot:** chat grounded in the farmer's own fields (decisions, risks, impact and open recommendations), with voice input and read-aloud through the browser's speech engine. If no AI is reachable, rule-based answers still work.

Sustainability, Field Trials and Testing & results

- **Sustainability:** the savings ledger (water, energy, CO₂, rupees, urea) with a monthly chart, a sustainability score (water, fertilizer cut, and share of savings measured by meters), each entry's method (measured, estimated, soil test), a hash-chain integrity check, and CSV/PDF export.
- **Field Trials:** pair a smart field with a control field following the usual practice; compare litres and kWh per acre and yield. Control plots are never counted as savings.
- **Testing & results:** the results of chapter 12, live in the app, with the improvements log.

Agronomist Desk, Settings and notifications

- **Agronomist Desk:** joined farms, risk meters per field, and notes to farmers.
- **Settings:** profile, language, alerts (in-app, browser push, SMS and WhatsApp through Twilio for urgent alerts, daily briefing), install as an app (PWA), and AI provider status.
- **Landing page:** live Open-Meteo soil moisture and rain for a real farm near Nashik, the problem, features, how it works, the field node and its parts list, and an impact calculator with cited assumptions.

4 How the decisions are made

Irrigation (backend/src/services/engine.service.ts)

Each field is analysed every 15 minutes, and after new telemetry. The engine uses FAO Irrigation and Drainage Paper 56 (Allen et al., 1998):

- **Crop stage:** from the planting date and the crop's four stage lengths; K_c is interpolated between $K_{c_{ini}}$, $K_{c_{mid}}$ and $K_{c_{end}}$, and roots deepen from their minimum to maximum depth over the first two stages.
- **Soil water:** $TAW = 1000 \times (\theta_{FC} - \theta_{WP}) \times Z_r$; $RAW = p \times TAW$; refill point = $\theta_{FC} - p \times (\theta_{FC} - \theta_{WP})$. Example: wheat at day 60 on loam gives TAW 168 mm, RAW 92.4 mm and a refill point of 17.3%.
- **Crop water use:** $ET_c = K_c \times ET_0$ (ET_0 from Open-Meteo, FAO Penman–Monteith).
- **Plan:** net depth = the deficit, capped per event (flood 50 mm, sprinkler 25 mm, drip 12 mm); gross = net ÷ efficiency (flood 55%, CWC studies; sprinkler 75% and drip 90%, FAO Training Manual 4); litres = gross × area; pump time = litres ÷ pump flow.

Order	Rule	Action
1	No reading, or the latest is older than 180 minutes	NO_DATA
2	Tank sensor fitted and level below its dry-run limit (default 15%)	BLOCKED_TANK
3	Soil moisture above the refill point	SKIP_WET
4	Not critical, and next-24 h rain $\geq \max(5 \text{ mm, half the deficit})$ with a $\geq 60\%$ chance	SKIP_RAIN
5	Farm has solar, field prefers solar, not critical, and the solar window starts within 6 h	WAIT_SOLAR
6	Smart schedules exist, none is open now, not critical	OUTSIDE_WINDOW
7	Otherwise	IRRIGATE (litres, minutes)

Critical means soil moisture $\leq \theta_{WP} + 0.35 \times (\text{refill} - \theta_{WP})$; a critical crop is watered even if rain is forecast. The **solar window** is the first run of forecast hours where predicted PV output (capacity × irradiance × 0.8 ÷ 1000) covers the pump's kW.

Risk scores

- **Disease risk** = humid hours (air humidity $\geq 90\%$) in the last day × 6 × a temperature factor (1 at 15–30 °C, 0.6 at 30–35 °C, 0.3 otherwise) + forecast humid hours × 2, plus up to 35 points from a recent leaf-scan finding.
- **Weather risk:** +25 for 35 °C or +50 for 38 °C in the next 3 days; +20 for a 25 mm or +45 for a 50 mm rain day; +20 for wind of 40 km/h or more.
- **Nutrient stress** from Soil Health Card ratings of N, P and K.
- **Crop health** = 100 – (35% water stress + 30% disease + 20% nutrient + 15% weather), blended 70/30 with an NDVI score when a recent satellite scene exists. Levels: Low < 35 ≤ Moderate < 65 ≤ High.

Fertilizer (fertilizer.service.ts)

- Soil ratings from the Methods Manual: Soil Testing in India (DAC 2011): N low below 280 kg/ha and high above 560; P 10 / 24.6; K 108 / 280.

- Dose: the crop's recommended N:P:K, +25% on low and –25% on high soils (PAU Package of Practices). Products use the Fertiliser (Control) Order 1985 minimum contents: urea 46% N, DAP 18% N and 46% P₂O₅, MOP 60% K₂O.
- Nitrogen split 50/25/25 at sowing and the two top dressings; notes for low organic carbon (< 0.5%), acidic (< 5.5) and alkaline (> 8.5) soils.

Soil-moisture forecast (forecast.service.ts)

Each field learns how its own soil moisture changes over 6-hour steps: $\Delta\theta = b_0 + b_1 \cdot \text{ETc} + b_2 \cdot \text{rain} + b_3 \cdot \text{irrigation}$, fitted by ridge least squares. The fitted model is used only with at least 16 samples, $R^2 \geq 0.2$, a negative ETc coefficient and a non-negative irrigation coefficient; otherwise the FAO-56 physics model is used. It is rolled forward hourly over the weather forecast, and a "needs water in N hours" alert goes out when the refill point is 3–48 hours away.

Satellite greenness

Sentinel-2 L2A scenes are found through the Element84 Earth Search STAC API; red and near-infrared bands are read only inside the field polygon, clouds and shadows are masked with the scene classification layer, and NDVI = $(\text{NIR} - \text{Red}) \div (\text{NIR} + \text{Red})$ is summarised as a mean, a range and zone shares.

The savings ledger

- Savings are counted only for periods with recorded water use (an irrigation event or a pump node reporting). Without evidence, a field claims no saving; control plots are never counted.
- Energy uses metered kWh per litre when an energy meter is fitted, else the pump rating, else a 30 m-head estimate (0.27 kWh per 1,000 L at 30% efficiency, BEE AgDSM); CO₂ uses 0.675 kg/kWh (CEA CO₂ Baseline Database v22.0); rupees use the farm's tariff.
- Each entry stores the hash of the previous entry, so any later edit breaks the chain and shows up in the integrity check.

5 AI in AgriGuard

The rule: the math decides, AI explains. Every number comes from the formulas and verified sources above. AI is used to see (photos), to read (Soil Health Cards) and to explain in the farmer's language. It is told to use only the given facts, never to add a number, product or dose, and to write numbers exactly as given.

Feature	Model / method
Leaf disease detection	MobileNetV2 fine-tuned on PlantVillage (38 classes), onnx-community/mobilenet_v2_1.0_224-plant-disease-identification-ONNX , fp32, run on the server with ONNX Runtime
Scan second opinion, pest mode, AI damage estimate	Gemini vision (chain below)
Soil Health Card reader	Gemini vision, JSON output of printed values only
"Why?" explanations (irrigation, risk, fertilizer, NDVI, disease advice)	Gemini, from a facts list built on the server
Daily briefing, weekly report, recommendation explain and translate, copilot, impact summary	Gemini, with rule-based fallbacks where possible
72-hour soil-moisture forecast and "needs water in N hours" alerts	Machine learning per field: ridge regression on the field's own history (moisture change per 6 hours from ETc, rain and irrigation); FAO-56 physics when there are fewer than 16 samples or the fit is not plausible (R^2 below 0.2 or wrong signs)
Leaf damaged area	No AI: pixel colour measurement (HSV)
Irrigation, risks, fertilizer, NDVI, ledger, pump alerts	No AI: formulas and rules (chapter 4)
Voice	The browser's Web Speech API

Provider chain and free-tier handling (backend/src/lib/ai)

1. **Gemini Flash**: every stable, versioned Flash model the key can use, newest first (currently 3.8, 3.7, 3.6, 3.5).
 2. **Gemini Flash-Lite** models (3.5, 3.1).
 3. Groq and OpenRouter, when keys are set.
 4. **Ollama** on the machine (Qwen2.5-VL 7B) for offline use.
 5. Rules.
- Each Gemini model has its own free-tier quota (20 requests a day for Gemini 3.8 Flash on this key). A model that hits its daily limit is paused until the quota resets at midnight Pacific; a per-minute limit or overload pauses it for 60 seconds; a retired model (HTTP 404) for a day.
 - Hidden reasoning is switched off where supported, and every request keeps 2,048 tokens of headroom, because some models ignore that setting. An answer that does not finish normally is rejected, so a cut-off answer is never shown or cached.
 - Answers are cached in the database by a hash of the prompt and facts, so repeat views cost nothing.

6 Data sources and verification

AgriGuard follows one rule: every number is either measured, taken from a verifiable source, or clearly labelled as a model estimate or a simulation. Nothing is made up.

What	Source
Weather forecast, ET ₀ , modelled soil moisture	Open-Meteo forecast API (no key); historical weather from the Open-Meteo archive (ERA5-Seamless), CC BY 4.0
Crop greenness	Copernicus Sentinel-2 L2A via Element84 Earth Search STAC
Crop coefficients, stages, root depth, depletion, soil water	FAO-56 (Allen et al., 1998), Tables 11, 12 and 19
Irrigation efficiency	CWC water-use-efficiency studies (flood 55%, Lok Sabha reply, 7 Dec 2023); FAO Training Manual 4 (sprinkler 75%, drip 90%)
Soil ratings	Methods Manual: Soil Testing in India (DAC 2011)
Fertilizer doses and nutrient contents	PAU Package of Practices; ICAR; Fertiliser (Control) Order 1985; prices from PIB releases
Disease advice (38 entries, 6 languages)	TNAU Agritech, ICAR institutes, NIPHM, CIB&RC major uses of pesticides, university extension services
Pump energy and CO ₂	BEE AgDSM (existing pumps 25–30% efficient); CEA CO ₂ Baseline Database v22.0 (0.675 kg/kWh)
Electricity tariffs	MSEDCL, PSPCL, MPERC and APERC tariff orders
Problem statistics	CGWB 2025; CEA General Review 2024; Agriculture Census 2015–16; ICAR; FAO
Model validation	PlantDoc (Singh et al., CoDS-COMAD 2020), CC BY 4.0

7 The field node (hardware)

A solar-powered ESP32 box that measures the field, posts JSON to the API and switches the pump starter on the server's command, with local safety cut-offs. Firmware, wiring, calibration and the full parts list are in [firmware/agriguard-node/](#).

Part	Measures / does	Approx. ₹ (2026)
ESP32 DevKit V1	Controller, Wi-Fi	350–450
Capacitive soil moisture v1.2/v2.0	Root-zone moisture	60–120
DS18B20 probe	Soil temperature	120–200
DHT22 (or SHT31)	Air temperature and humidity	250–350
RS485 NPK sensor (optional)	Nitrogen, phosphorus, potassium	3,500–6,500
YF-S201 flow sensor	Litres pumped	250–400
PZEM-004T (optional)	Pump energy (kWh)	1,000–1,500
JSN-SR04T ultrasonic	Tank level	350–550
INA219 (optional)	Solar input	150–250
Relay, 6 W solar panel, MPPT charger, 18650 cells, box, wiring	Pump switching and off-grid power	≈ 1,500–3,000

Totals: basic node about ₹4,000–4,500; full node with NPK, energy meter and solar monitor about ₹10,000 (most of it the NPK sensor).

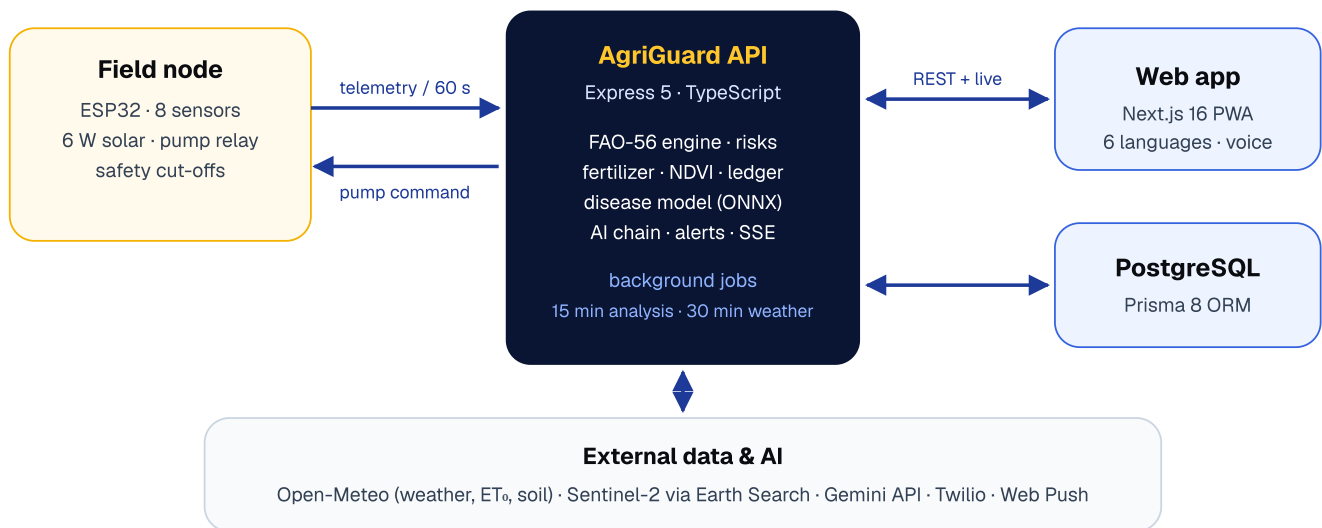
Telemetry protocol

```
POST /api/device/telemetry      header  X-Device-Key: <key from the Devices tab>
{ "soilMoisture": 27.4, "soilTemp": 24.1, "airTemp": 31.2, "humidity": 58,
  "flowTotalL": 18234, "flowRateLpm": 42, "energyTotalKwh": 311.2, "powerW": 2940,
  "tankLevel": 64, "solarW": 18.5, "batteryPct": 91, "rssi": -67, "pumpOn": true }

200 → { "pump": "ON" | "OFF", "runSeconds": 900, "reason": "...",
        "reportEverySeconds": 60 (15 while pumping), "serverTime": "..." }
```

All fields are optional and range-checked on the server. Flow and energy totals are cumulative, so the server turns differences into litres and kWh per irrigation event. The server flags "no flow while the pump is on" and "flow while the pump is off" (a leak), marks nodes offline after missed reports, and detects stuck sensors (identical readings for 6 hours).

8 Architecture



- **Frontend:** Next.js 16.3 (App Router, React 19.2, Tailwind), TanStack Query, Leaflet maps, Recharts; installable PWA with a service worker; runs on port 3100 in development.
- **Backend:** Express 5 on Node 24 (port 5000): zod-validated routes, services, and a scheduler inside the same process; live updates over Server-Sent Events (`/api/stream`); uploads served from `/uploads` .
- **Database:** PostgreSQL through Prisma 8 (release candidate), with contract-first migrations; an embedded Postgres for local development.
- **Background jobs:** Open-Meteo sync every 30 min; field analysis every 15 min; device health every 5 min; hourly ledger sync, stuck-sensor and forecast alerts; briefing checks every 10 min.

9 Code map

backend/

Path	Contents
<code>src/app.ts</code> , <code>src/server.ts</code>	Express app (helmet, CORS, rate limits, routes, trust proxy) and startup (model preload, push keys, jobs)
<code>src/config/</code>	<code>env.ts</code> (all settings), <code>database.ts</code> , <code>paths.ts</code>
<code>src/routes/</code>	auth, farm, field-ops, device (telemetry and device management), platform (everything else)
<code>src/services/</code>	engine (decisions, risks), agronomy (FAO-56 maths), weather (forecast, solar window), open-meteo-soil, forecast, ndvi, scan, fertilizer, recommendation, explain, ai-features, copilot, ledger, impact, trial, advisor, telemetry, pump, device, notification, dashboard, overview, validation, access, auth
<code>src/lib/</code>	<code>ai/</code> (providers, chain, cache), <code>plant-model.ts</code> , <code>leaf-analysis.ts</code> , <code>ndvi.ts</code> , <code>events.ts</code> (SSE), <code>push.ts</code> , <code>twilio.ts</code> , <code>field-lock.ts</code> , <code>time.ts</code> , <code>geo.ts</code>
<code>src/data/</code>	<code>crops.ts</code> (FAO-56 profiles and doses), <code>soils.ts</code> (soil limits, efficiencies, energy, CO ₂), <code>disease-knowledge.ts</code> (38 entries × 6 languages)
<code>src/i18n/</code>	Server-side messages, crop and soil names in 6 languages
<code>src/jobs/scheduler.ts</code>	Background jobs
<code>prisma/</code> , <code>migrations/</code>	Schema contract and 3 migrations (baseline, live farm features, Open-Meteo source) with <code>db</code> and <code>production</code> refs
<code>scripts/</code>	<code>seed-demo.ts</code> , <code>simulate-device.ts</code> , <code>db.mjs</code> (local Postgres), <code>validate-disease-model.ts</code> , <code>replay-season.ts</code> , <code>test-report.ts</code>
<code>test/</code>	Unit tests: agronomy, engine, weather, fertilizer, knowledge base, AI rate limits
<code>data/validation/</code>	Results files for the Testing & results page

frontend/ and firmware/

Path	Contents
<code>src/app/</code>	Routes: landing, login, register, and the dashboard group (dashboard, farms, fields/[id], scan, risk, recommendations, copilot, sustainability, trials, validation, advisor, settings)
<code>src/components/</code>	Feature folders: ai (explain), dashboard, field (and ops), scan, recommendations, copilot, impact, trials, validation, advisor, settings, landing, layout, ui
<code>src/i18n/</code>	Provider and message catalogue: 17 namespaces × 6 languages, typed keys
<code>src/lib/</code>	<code>api.ts</code> (typed client), <code>types.ts</code> , <code>format.ts</code> (Indian number formats), <code>impact.ts</code> (calculator constants)
<code>firmware/agriguard-node/</code>	Arduino sketch, config template and a complete build guide (Render Blueprint: <code>render.yaml</code> at the root)

10 API reference

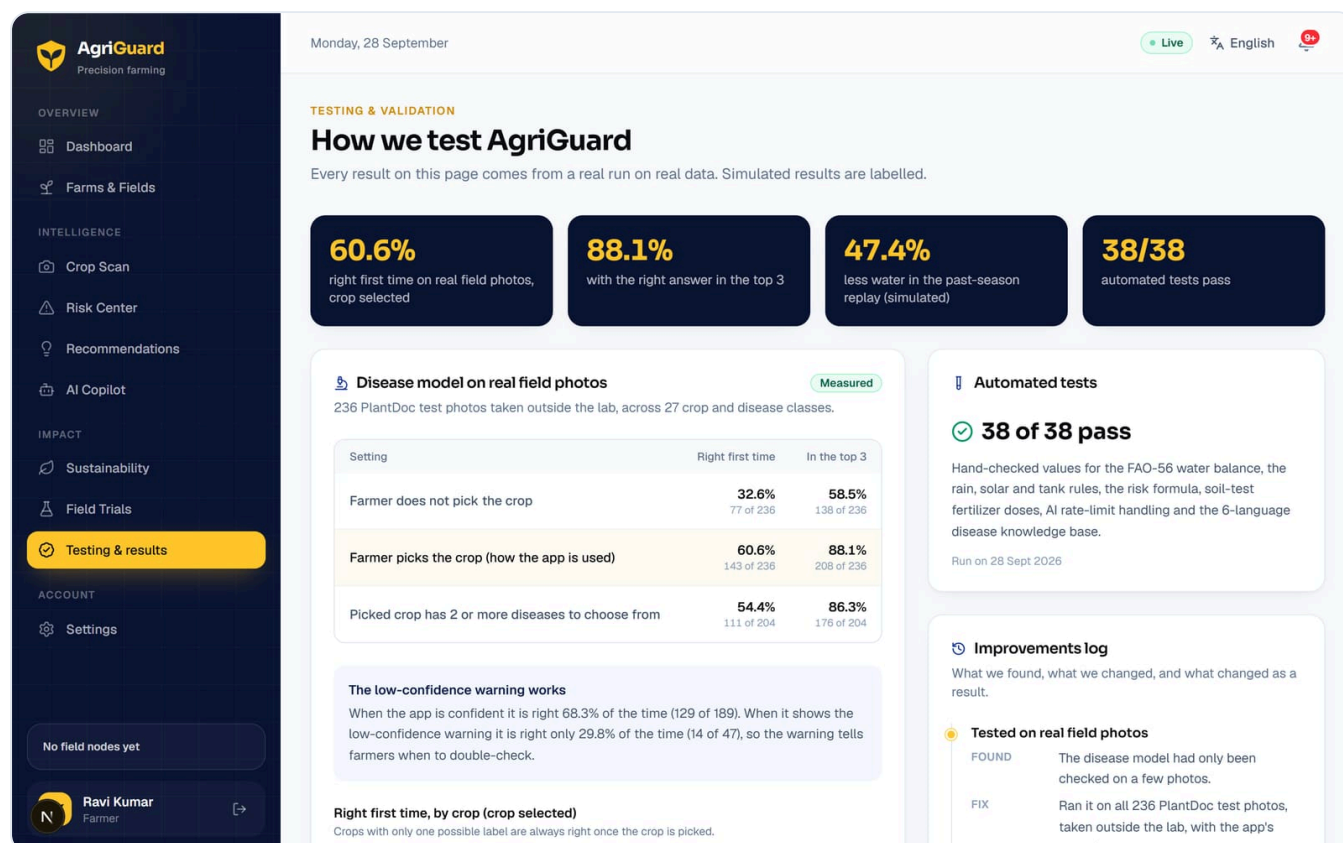
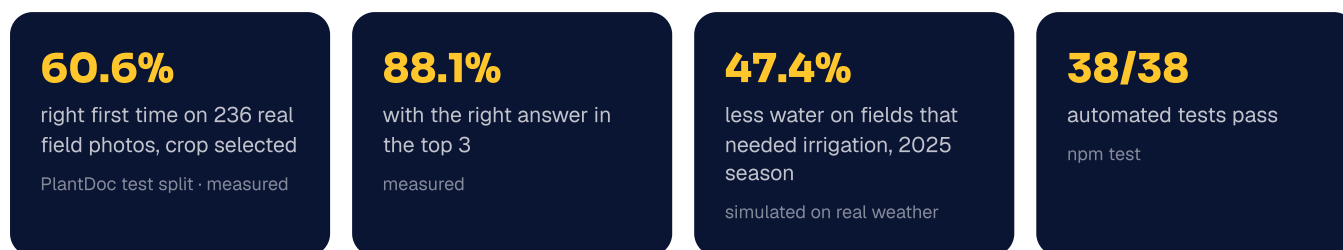
All paths start with `/api`. Everything except auth, telemetry, `/meta/crops`, `/push/public-key`, `/validation` and `/health` needs `Authorization: Bearer <token>`. Clients send `X-Language` for localised answers.

Area	Endpoints
Auth	<code>POST /auth/register</code> · <code>POST /auth/login</code> · <code>GET /auth/me</code> · <code>PATCH /auth/me</code>
Farms	<code>GET POST /farms</code> · <code>GET PATCH DELETE /farms/:farmId</code> · <code>GET POST /farms/:farmId/fields</code> · <code>GET PATCH DELETE /farms/:farmId/fields/:fieldId</code> · <code>GET /farms/:farmId/weather</code>
Field operations	<code>GET /fields/:id/overview</code> · <code>PATCH /fields/:id</code> · <code>POST /fields/:id/analyze</code> · <code>GET POST /fields/:id/observations</code> · <code>GET /fields/:id/observations/raw</code> · <code>GET /fields/:id/forecast</code> · <code>GET POST /fields/:id/crops</code> · <code>PATCH /fields/:id/crops/:cropId/harvest</code>
Pump and schedules	<code>POST /fields/:id/pump</code> · <code>GET /fields/:id/events</code> · <code>GET /fields/:id/decisions</code> · <code>GET POST /fields/:id/schedules</code> · <code>PATCH DELETE /fields/:id/schedules/:scheduleId</code>
Devices	<code>POST /device/telemetry</code> (X-Device-Key) · <code>GET POST /fields/:id/devices</code> · <code>GET /devices</code> · <code>PATCH DELETE /devices/:id</code> · <code>GET /devices/:id/key</code> · <code>POST /devices/:id/rotate-key</code>
Fertilizer	<code>GET POST /fields/:id/fertilizer-plans</code> · <code>POST /fields/:id/fertilizer/preview</code> · <code>POST /fields/:id/fertilizer-plans/:planId/applied</code> · <code>POST /fields/:id/soil-card</code> (photo)
Satellite and reports	<code>GET POST /fields/:id/ndvi</code> · <code>GET /fields/:id/report</code> · <code>GET /fields/:id/explain/:topic</code> (irrigation, risk, fertilizer, ndvi) · <code>POST /fields/:id/notes</code>
Scans	<code>GET /scans/crops</code> · <code>POST /scans</code> (photo) · <code>GET /scans</code> · <code>GET /scans/:id</code> · <code>GET /scans/:id/explain</code>
Insights	<code>GET /dashboard</code> · <code>GET /recommendations</code> · <code>PATCH /recommendations/:id</code> · <code>GET /recommendations/:id/explain</code> · <code>GET /impact</code> · <code>GET /impact/report.csv</code> · <code>GET /impact/report.pdf</code>
AI	<code>POST /copilot/chat</code> · <code>GET DELETE /copilot/history</code> · <code>GET /ai/status</code> · <code>POST /ai/translate</code> · <code>GET /ai/briefing</code>
Trials and advisors	<code>GET POST /farms/:farmId/trials</code> · <code>PATCH DELETE /trials/:id</code> · <code>POST /farms/:farmId/share</code> · <code>POST /farms/:farmId/share/reset</code> · <code>GET /farms/:farmId/advisors</code> · <code>DELETE /farms/:farmId/advisors/:advisorId</code> · <code>POST /advisor/join</code> · <code>GET /advisor/overview</code>
Notifications and meta	<code>GET /stream</code> (SSE) · <code>GET /notifications</code> · <code>POST /notifications/read</code> · <code>GET /notifications/channels</code> · <code>POST /notifications/test</code> · <code>POST /push/subscribe unsubscribe</code> · <code>GET /push/public-key</code> · <code>GET /meta/crops</code> · <code>GET /meta/crop-profiles</code> · <code>GET /geocode</code> · <code>GET /validation</code> · <code>GET /health</code>

11 Data model

Table	Holds
User	Account, role, language, alert preferences
Farm, FarmAdvisor	Farm location, irrigation method, tariff, solar capacity, usual practice, share code; agronomists joined to a farm
Field, Crop	Area, soil, boundary, pump, refill point; crops with planting and harvest dates
Device, FieldObservation	Field nodes, keys, pump mode and sensors fitted; every reading with its source (device, manual, simulator, Open-Meteo)
IrrigationEvent, IrrigationDecision, PumpSchedule	Pump runs with litres and kWh (measured or estimated); each decision and its reason; smart schedule windows
IrrigationPrediction, WeatherPrediction, NutrientPrediction	History of each analysis run's predictions
AIAssessment, CropImage, DiseasePrediction	Risk assessments and scans, scan photos, ranked disease predictions
Recommendation	Prioritised actions with evidence, status and source
FertilizerPlan, NdviSnapshot	Soil-test plans vs blanket dose; satellite scenes with NDVI statistics, zones and overlay image
SustainabilityRecord	The hash-chained savings ledger
Trial	Treatment vs control field pairs and yields
Notification, PushSubscription, ChatMessage, AiCache	Alerts with per-channel status, browser push endpoints, copilot history, cached AI answers
ProcessingJob	Reserved; not written by the current code

12 Testing and validation results



The Testing & results page in the app shows these results live, in all six languages, with the improvements log.

Disease model on real field photos MEASURED

The on-device model was trained on PlantVillage lab photos (95.41% on its own test split). We ran it with the app's own steps on the full PlantDoc test split (236 photos, 27 classes), images taken outside the lab.

Setting	Right first time	In the top 3
Farmer does not pick the crop	32.6% (77/236)	58.5% (138/236)
Farmer picks the crop (how the app is used)	60.6% (143/236)	88.1% (208/236)
Picked crop has 2+ diseases to choose from	54.4% (111/204)	86.3% (176/204)

- The low-confidence warning works: confident answers are right 68.3% of the time (129/189), flagged ones 29.8% (14/47).
- For comparison, the PlantDoc paper reports 15.08% for a PlantVillage-trained VGG16 on uncropped PlantDoc photos.

- Weakest: tomato (25/69 with the crop picked) and corn blight/rust, which are often read as gray leaf spot. 15 test photos have file names that contradict their label; they stay in the numbers.
- Gemini's accuracy on the same photos has not been scored yet: the free tier allows 20 requests a day per model.

Past-season replay SIMULATED

Real 2025 weather (Open-Meteo ERA5-Seamless) for each demo field's real location, run day by day through a FAO-56 water balance. AgriGuard's actual decision function runs each morning; all 1,540 decisions were re-derived independently with zero mismatches. The baseline is a **rain-blind weekly schedule** that replaces the crop's full FAO-56 water use every 7 days. It contains no guessed "farmer practice" numbers.

Field (crop, method)	AgriGuard (million L)	Weekly schedule	Saved	Stress days
North Field (tomato, drip)	3.76	7.77	51.6%	0 / 0
Canal Block (paddy, flood)	17.28	28.92	40.2%	2 / 17
Ganna Plot (sugarcane, drip)	13.08	27.58	52.6%	0 / 0
Kanda Plot (onion, sprinkler)	3.25	5.22	37.8%	1 / 21
Chilli Block (chilli, drip)	1.25	3.99	68.5%	1 / 3
Conservative total (5 fields)	38.63	73.48	47.4%	4 / 41

Saved on those 5 fields: **34.85 million litres, 8,677 kWh, 5,857 kg CO₂ and ₹40,418** at each farm's tariff. The crops still got 99.99% of the water they needed. Four more fields (soybean, maize, cotton) needed no irrigation in 2025 because rain kept the soil above the refill point; with them, the total is 64.0%, but they are kept out of the headline.

Assumptions: observed rain is used as the forecast (flatters AgriGuard slightly); the weekly schedule ignores rain, so a farmer who watches the rain would save less; paddy ponding and percolation are not modelled; each season starts at field capacity; field sizes and pumps are demo values.

Automated tests

38 unit tests with hand-checked values: crop stages and Kc; TAW, RAW and refill points; irrigation plans per method; every decision rule (no data, tank, wet soil, rain skip and its probability and critical overrides, solar wait, schedules); the risk formula and NDVI blend; forecast summaries and the solar window; soil ratings, doses, splits and notes; soil-name resolution; knowledge-base completeness in all 6 languages; AI rate-limit pauses. Run `npm test` ; `npm run test:report` writes the counts shown in the app.

Improvements log (highlights)

Found	Fix	Result
The disease model had been checked on only a few photos	Full PlantDoc test split with the app's own steps	60.6% right first time with the crop picked, vs a 15.08% published baseline
First replay claimed 79% less water against an unverified baseline that ignored rain	Rain-blind FAO-56 weekly baseline; 79% dropped	47.4% conservative headline

Found	Fix	Result
"Red (sandy loam)" soil was read as sandy	Exact name matching and a regression test	Chilli field uses red-soil limits (FC 19%, WP 8%)
Free Gemini quota (20/day) ran out during testing	Rotation through every available Flash model	The next model answered, correctly in Marathi
Some models ignored "no hidden reasoning" and cut answers short	Token headroom; cut-off answers rejected	The briefing returns 3 complete bullets
Website said the node costs ₹2,150 and reports every 15 min	Matched to the parts list and firmware	₹4,000–4,500; every 60 s (15 s while pumping)
Behind a proxy all visitors shared one IP for rate limits	Trust exactly one proxy	Found and fixed in a deploy rehearsal
Disease advice had outdated pathogen names and unsourced sprays	All 38 entries checked against TNAU, ICAR, NIPHM, CIB&RC	9 names updated, sprays changed for 15 diseases, 13 lines fixed, 5 practices added

13 Running, seeding and deploying

Local development

```
# Backend (Node 24)
cd backend
npm install
npm run db           # local PostgreSQL in .pgdata (keep running)
npx prisma db migrate --to production
npm run seed:demo    # 6 demo accounts with real weather and satellite data
npm run dev          # http://localhost:5000

# Frontend
cd frontend
npm install
npm run dev -- -p 3100 # http://localhost:3100

# Checks
npm test             # backend unit tests
npx tsc --noEmit -p . # type checks (both apps)
```

Environment settings (backend/.env)

Variable	Purpose
<code>DATABASE_URL</code> , <code>JWT_SECRET</code>	Required
<code>GEMINI_API_KEY</code> (+ optional <code>GEMINI_MODEL</code>)	AI; <code>GROQ_*</code> , <code>OPENROUTER_*</code> , <code>OLLAMA_URL/MODEL</code> are optional extra providers
<code>FRONTEND_URL</code> , <code>PUBLIC_API_URL</code> , <code>APP_TIMEZONE</code>	Links, node setup URL, day boundaries (default Asia/Kolkata)
<code>TRUST_PROXY</code>	Number of proxies in front of the API (1 on Render)
<code>TWILIO_*</code> , <code>VAPID_*</code>	SMS/WhatsApp and browser push (push keys are generated if empty)
<code>ENABLE_JOBS</code> , <code>PLANT_MODEL_DTYPE</code>	Background jobs on/off; model precision (fp32 by default)

The frontend needs `NEXT_PUBLIC_API_URL` (and `NEXT_PUBLIC_SITE_URL` for links) at build time. API keys live only in the backend.

Render deployment

1. Dashboard → New → Blueprint → connect `marvillage/AgriGuard` , branch `main` .
2. Render creates `agriguard-db` (PostgreSQL), `mavestorm-agriguard-api` and `mavestorm-agriguard` (free plan, Singapore); enter `GEMINI_API_KEY` .
3. The API's start command applies pending migrations (`prisma db migrate --to production` , a no-op when current) and starts the server; `/api/health` is the health check. `JWT_SECRET` is generated by Render.
4. Load demo data once from a workstation: `DATABASE_URL=<external URL> npm run seed:demo` .
5. Every push to `main` redeploys automatically.

Free plan: services sleep after 15 minutes idle (first request about a minute), uploaded photos are lost on redeploy, and the free database has an expiry date.

14 Security and reliability

- **Authentication:** passwords hashed with bcrypt; JWT bearer tokens valid for 7 days; the token lives in the browser's storage and is sent as a header, not a cookie.
- **Authorisation:** every farm and field request goes through `farmAccess` / `fieldAccess` (owner, joined advisor with read access, or admin); writes need owner rights.
- **Devices:** each node has its own key (`X-Device-Key`), masked in lists, viewable by the farm owner and rotatable at any time; telemetry values are range-checked.
- **Input:** zod schemas on request bodies; uploads limited to 10–12 MB and images only.
- **Transport and headers:** helmet security headers, CORS, rate limits (100 login/register attempts and 2,000–5,000 other requests per 15 minutes per IP), with `trust proxy` set so limits apply per visitor behind Render.
- **Secrets:** keys only in backend environment variables; nothing secret in the repository (checked before every push).
- **Reliability:** per-field locks prevent overlapping analyses; the AI chain falls back model by model to rules; background jobs are wrapped so one failure never stops the others; migrations are idempotent on restart.

15 Limitations and roadmap

Known limitations

- **No measured savings yet.** The ledger stays at 0 until a field node records real pump runs. The next step is a 1–2 week smart-plot vs control-plot trial with the node.
- **Default usual-practice baseline** for live savings (flood 70 mm every 7 days, sprinkler 35 mm every 4, drip 8 mm daily) is not yet verified against a published source; farms should enter their own practice.
- **Disease model:** 60.6% right first time on real field photos, weaker for tomato and corn; the second opinion and the low-confidence flag help, and Gemini's accuracy still needs scoring.
- **Free Gemini tier:** about 20 requests a day per model; heavy use needs a paid key.
- **Replay:** simulation with a perfect forecast and no paddy ponding; not a substitute for a field trial.
- **Hosting:** free plan sleep, ephemeral uploads, database expiry.

Roadmap

- Field trial with a built node (measured litres, kWh, ₹ and CO₂ in the ledger).
- Diesel pump support, adding a "fuel saved" metric.
- LoRa uplink for farms without Wi-Fi; offline-first sync.
- Score Gemini's second opinion on PlantDoc; retrain the disease model on field photos.
- Pilots with KVKs and FPOs; subscription model for FPO and agronomist dashboards.

AgriGuard · Team Mavestorm. Code: github.com/marvillage/AgriGuard · Every drop measured. Every field protected.